

H1 MCQ Answers

1. a
2. b
3. c
4. c
5. d
6. d
7. a
8. d
9. c
10. c
11. c
12. c
13. b
14. c
15. b
16. d
17. c
18. d
19. c
20. c
21. c
22. c
23. d
24. d
25. c
26. d
27. c
28. c
29. b
30. b

Question 1

Gastric proton/potassium-ATPase (H^+/K -ATPase) is the major enzyme involved in gastric acid secretion in the stomach. Recognizing the calming effects that tea can have on an upset stomach, biochemists quickly showed that the tannic acid found in tea is a potent inhibitor of this enzyme. Gastric proton/potassium-ATPase (H^+/K -ATPase) has two substrates, potassium ion (K^+) and ATP. (K_m is substrate concentration at $\frac{1}{2} v_{max}$)

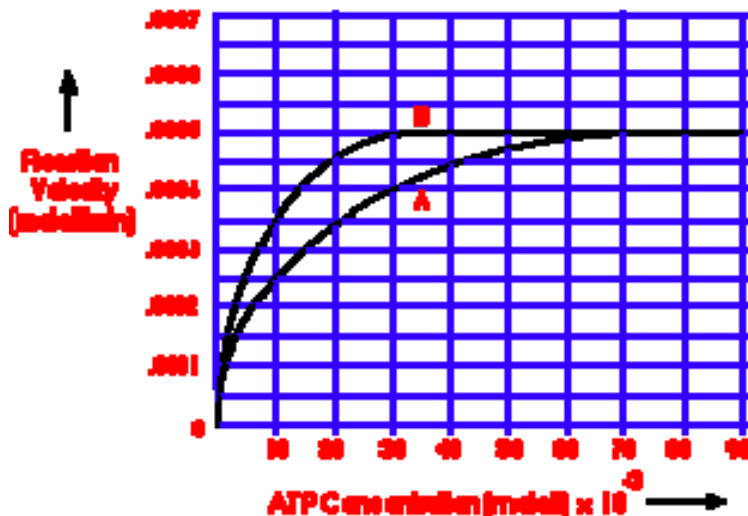


Figure 1. Reaction velocity-substrate concentration curves for Gastric proton/potassium-ATPase (H^+/K -ATPase) with unlimiting levels of K^+ but varying levels of ATP. A. with tannic acid; B. without tannic acid.

a) What is the minimum ATP concentration needed to saturate the enzyme in this system without tannic acid? [1]

30 mole/l x 10⁻³

b) By comparing k_m values based on the graphs in figure 1, explain which treatment would result in the higher affinity of the enzyme for the substrate? [2]

B

K_m value for B was 5mole/l x 10⁻³

while A was 10 mole/l x 10⁻³

V_{max} would be reached at lower substrate concentrations

c) Name the type of inhibition that is involved above and provide evidences for your choice [2]

At very high substrate concentration

V_{max} can be reached/very close to values without inhibitor added

Inhibition can be reversed/competitive inhibition

Compete with substrate for active site

d) With reference to the structure of the active site. suggest its possible mechanism of inhibition and why the inhibitor cannot be broken down by the enzyme [2]

Inhibitor is complementary in shape

To contact residues of the active site

Binding is reversible

Therefore, another substrate molecule is able to compete with the inhibitor at the active site

Catalytic residues unable to act on the bonds of the inhibitor,

Therefore cannot be broken down.

e) Describe two ways in which enzymes interact with substrates to specifically increase the likelihood of a reaction [2]

Substrate binding to active site results in physical stress in bonds

Increases the likelihood of bonds breaking

R groups of enzymes can change the charge on the substrate/alter the distribution of electrons/other changes

Increase the reactivity of the substrate

g) Giving an example in each, one main difference between prosthetic groups and coenzymes that are involved in enzymatic reactions [2]

Prosthetic groups – tightly bonded to enzyme; Coenzyme – loosely bonded/not bonded when reaction has not occurred

Prosthetic group: haeme group

Coenzyme: FAD, NADH

Question 2

A cross was made between a pure breeding round-leafed, tall plant and a oval-leafed, dwarf plant produced all round-leafed, tall plants.

(i) Using this information determine which alleles are dominant and which are recessive.

Round -leaf is dominant allele, oval-leaf is recessive allele.

Tallness is dominant allele while dwarfness is recessive allele

[2]

In **another** cross between a round-leafed, tall plant and a round-leafed dwarf plant following offspring were produced:

121 round-leafed, tall plants

124 round-leafed, dwarf plants

42 oval-leafed, tall plants

37 oval-leafed, dwarf plants

Using appropriate symbols, deduce the genotypes of the 2 parents and explain how you arrive at your answer

(ii) Symbols:

R - be the dominant allele for round leaf
r – be the recessive allele for oval leaf
T – be the dominant allele for tall
t – be the recessive allele for dwarf

[1]

(iii) Genotypes of the 2 parents

RrTt

Rrtt

[2]

(iv) Explanation:

To get an oval-leaved (rr) plant , each parent must contribute a recessive allele (r).
Hence both parents are heterozygous at the R locus (Rr).

To get a dwarf plant (tt), each parent must contribute a recessive allele (t).

However, one parent is heterozygous at the T locus (Tt) and the other is homozygous recessive (tt)

in order to fulfill the ratio of 3:3:1:1.

[3]

[Total: 8 marks]

Question 3

- a) What are the main differences in the structure and organization of prokaryotic and eukaryotic genomes? [4]

At least 1 from structure and 1 from organization.

Structure

Area of Comparison	Feature	Prokaryotic Genome	Eukaryotic Genome
Genes & bases	*Number of genes and bases	Larger number of bases - larger number of genes / coding regions;	Smaller number of bases - smaller number of genes /coding regions;
Chromosomes	*Number of chromosomes	Single chromosomes - usually haploid (ie only 1 set of chromosomes);	Multiple chromosomes - usually in diploid or higher ploidy

			levels; (Two or more sets of chromosomes)
	*Chromosome Structure	Double-stranded and circular DNA.	Double-stranded and linear DNA.
	*Involvement of proteins in DNA packaging	Supercoiling and looping brought about by non-histone proteins , (which are positively charged).	-DNA wrapped around an octamer of basic histone proteins (also positively charged) to form nucleosomes, which are further coiled to form solenoids. -other scaffolding proteins (non-histone proteins) also involved in packaging.
	*Level of condensation of DNA	Association with histones / scaffolding proteins allow for increased structural complexity/folding to higher degree of condensation	No association with histones thus does not achieve same level of condensation complexity as eukaryote

Organisation

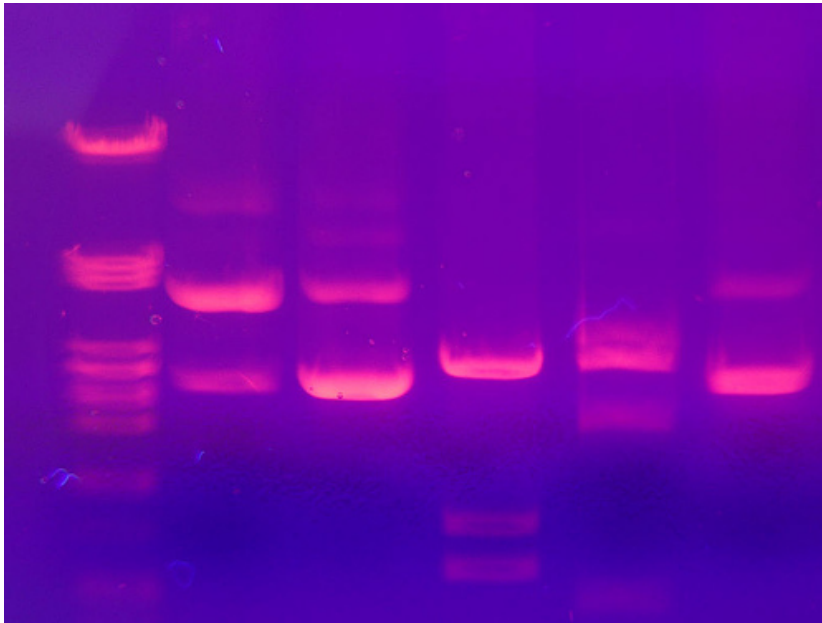
Area of Comparison	Feature	Prokaryotic Genome	Eukaryotic Genome
Organisation of Genomes	*Location of Functionally-Related Genes on Chromosomes	Genes that encode proteins for the same metabolic pathway are grouped together in an operon .	Genes that encode functionally related proteins are usually located on different chromosomes .
	*Control by Promoter	Genes grouped in an operon are controlled by a single promoter .	Each gene is under the control of their individual promoters .
	*Presence of Introns	Typically no introns in prokaryotic genome	Presence of introns (non-coding regions) in eukaryotic genome
	*Presence of Telomeres (can be under structure as well)	No telomeres since circular DNA allows complete replication of daughter DNA molecules.	Presence of telomeres , which are repetitive and non-coding sequences that are lost with successive rounds of

			replication as the chromosome shortens.
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b) Describe two functions of repetitive sequences. [2]

- ***They could be origins of replication where DNA polymerase binds to initiate replication.***
- ***They encode rRNA which are needed in many copies to form ribosomes.***
- ***They code for histone proteins which are needed in many copies for DNA packaging.***
- ***They could be centromeres which are the binding sites for kinetochore proteins for the movement of chromosomes.***
- ***They could be telomeres which maintain the integrity of the chromosome. [Any two]***

Question 4



Mark

A

B

C

D

E

Fig shows a gel electrophoresis result of dna obtained from similar looking flagella obtained from 5 species of marine bacteria. Species A to E were each retrieved from different oceans of the world.

ai) Name the two species that are most likely to evolve a flagella similar to A, as a result of convergent evolution. [1]

C and D

aii) Give four advantages of using such a method in the classification of the 5 organisms into the phylogenetic tree. [4]

Patterns of inheritance are straightforward, data more interpretable

Molecular traits that are similar are less likely due to convergence

The number of potential point mutation differences between species is likely to be very large, so even if there are independent identical mutations, those will be outweighed by the number of dissimilar mutations

The number of mutations can be used to measure time because we can calculate the rate of mutation/knowning the rate of mutation, we can use the number of mutations to calculate time organism has evolved from one species to another.

b) It was found that these bacteria populations all have evolved a similar genetic resistance against the toxins produced by another marine prokaryote *Vibrio marinus*. Suggest, based on Darwin's ideas of natural selection, how these populations could have evolved this particular characteristic. [3]

- **variations in any population**
- **Resistant and non-resistant bacteria**
- **Toxins exerts a selection pressure against individuals of the population**
- **Those with the resistance genes are selected for/have selective advantage**
- **Those with resistance genes are selected against/have selective disadvantage**
- **Differential survival and reproduction**
- **Resulting in more individuals with the genes conferring mutation**
- **Over generations, proportions of those at selective advantage increases.**

c) Outline the processes and principle in Agarose Gel Electrophoresis. [5]

- **Loading of dna in well**
- **Dye added to track movement**
- **Glycerol to help dna sink**
- **Electric field applied at opposite ends**
- **DNA, negatively charged**
- **Move towards the anode/positive electrode**
- **Larger fragments will move slower**
- **Fragments of different sizes will separate out as bands**
- **Ethidium bromide used to stain/intercalate DNA in gel**
- **UV light will make DNA fluoresce**

Question 5

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of Genomes	Functionally-Related Genes on Chromosomes	proteins for the same metabolic pathway are grouped together in an operon.	functionally related proteins are usually located on different chromosomes.
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Question 6

a) What are the advantages and disadvantages of the use of PCR? [4]

Advantages

- Amplification of small amounts of DNA can be done***
- time-efficient, many copies of DNA can be obtained within several hours***
- Not necessary to use living organisms like bacteria to replicate DNA***

Disadvantages

- Part of the gene sequence needs to be known for the design of primers***
- Sequences to be amplified must be short, up to 40kb***

- *If proofreading polymerases not used, higher frequency of copying errors*
- *limit to the number of accurate copies that can be made by PCR due to the accumulation of relatively copying errors*

b) Contrast the features and normal functions of totipotent, pluripotent and multipotent stem cells. [6]

Totipotency	Pluripotency	Multipotency
Potential of cells to develop into <u>any and all of the human cells including the extra-embryonic tissues</u> [1]	Potential of cells to <u>give rise to any differentiated cell types but not the extra-embryonic tissues.</u> [1]	Potential of cells that are <u>descendants of pluripotent stem cells and can give rise to specialized cells in particular tissues.</u> [1]
Early embryonic stem cells to give rise to all the cells of the both the embryo and the extra-embryonic tissues like the placenta. [1]	Embryonic stem cells found in the inner cell mass of embryo, give rise to all the cells that will eventually form the embryo – the endoderm, mesoderm and ectoderm. [1]	Adult stem cells Umbilical cord stem cells Haematopoietic stem cells 1. Maintain a stable stem cell population within the tissue 2. Generate cells that will differentiate into specialised cells to replace those lost to cell death / injury in the specific tissue. [1]

c) Outline the procedure to produce the human growth hormone (hGH) from bacterial cells. [10]

Isolate hGH gene from human DNA

- *using appropriate restriction enzymes to cut the hGH gene and use the same restriction enzyme*
- *to cut an expression vector*
- *producing complementary sticky ends*
- *which will anneal to each other via Hydrogen bonds*
- *Ligase is added*
- *to seal the nicks between the nucleotides / form phosphodiester bonds between the expression vector and the hGH gene*

- Thus forming a recombinant plasmid,
- although empty vector, circularized hGH gene may also result

Transformation

- Resultant DNA is introduced into bacteria via transformation
- through Calcium Chloride treatment followed by heat shock OR electroporation

Selection

- Transformed bacteria are plated onto agar plates
- and allowed to grow to form colonies

[Any One method – but max 3 marks only]

(Double selection)

- Bacteria colonies containing bacteria that have been successfully transformed by the recombinant DNA will be identified using the double selection method –
- They could be identified through replica plating
- with one plate containing ampicillin and the other containing tetracycline.
- where if the gene was inserted in the middle of the tetracycline gene / tetracycline gene disrupted by the inserted gene of interest,
- bacteria with recombinant plasmids would be resistant to ampicillin but not tetracycline. (or any other scenario which is correct)
- The colonies present on the ampicillin plate but not the tetracycline plate would be the ones with recombinant DNA. [1]

OR (β -galactosidase activity)

- Bacteria colonies containing bacteria that have been successfully transformed by the recombinant DNA will be identified using β -galactosidase activity –
- where if the gene was inserted in the middle of the β -galactosidase gene,
- bacteria with recombinant plasmids would not be able to express the enzyme.
- They could be identified through growing bacteria in media containing IPTG and X-gal,
- bacteria with recombinant plasmids not be able to break down X-gal and thus will appear white rather than blue [1]

OR (screening by probe)

- Some bacteria from each of the bacteria colonies on the agar plate could be transferred onto a nitrocellulose membrane
- and allowed to lyse to release their DNA which would bind onto the membrane
- A single stranded DNA probe complementary to part of the hGH gene sequence could be synthesized
- and allowed to bind with the denatured DNA on the membrane
- (correct visualization) allowing the identification of the positions of the colonies on the original plate with the recombinant DNA. [1]

- ***Bacteria from the colonies with the recombinant plasmid would be isolated***
- ***and allowed to grow in large numbers in biofermenters***
- ***Inducers would have to be added to induce the expression of the hGH protein***
- ***which would then be isolated, purified and processed for human use***

Question 7

a) Describe in detail the structure of DNA [12]

- **Two strands of polynucleotide.**
- **Arranged in double helix**
- **Antiparallel**
- **Sugar-phosphate backbones, with nitrogenous bases in the centre**
- **5' to 3', the other strand 3' to 5' direction**
- **Nucleotide made of pentose sugar**
- **Deoxyribose, 2nd carbon lacks oxygen molecule**
- **Phosphate group**
- **Base, either purine or pyrimidine**
- **consisting of adenine, guanine, cytosine or thymine**
- **Complementary base pairing between bases.**
- **A pairs with T and G pairs with C**
- **Ratio is 1:1 respectively**
- **Via hydrogen bonding**
- **Width between the two strands constant**
- **One turn 3.4nm long and ten base pair per turn**

[1 mark each]

b) Explain the main events of transcription. [8]

- **RNA polymerase binds to the promoter**
- **DNA helix unwinds**
- **Sense strand is transcribed**
- **Polymerase moves from 3' to 5' end of the sense strand**
- **Adding RNA, complementary to DNA bases**
- **New strand synthesized in 5' to 3' direction**
- **A pairs with U and G pairs with C**
- **RNA nucleotides joined by phosphodiester bonds**
- **mRNA released**
- **DNA rewinds**
- **mRNA spliced**
- **leaves nuclear membrane**

[1 mark each]